

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data integration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure B
Analytical Problem Solving

<i>Criteria</i>	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	20%	25%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO₂ - ATL-1

ANALYTICAL QUESTIONS

Sol:- Given Data (Sorted)

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25,
33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Number of observations (n) = 27

(a) Mean and Median

Mean = Sum of all values

$$13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22 + 22 + 25 + 25 + 25 + 33 + 35 + 35 + 35 + 35 + 36 + 40 + 45 + 46 + 52 + 70 = 813$$

$$\text{Mean} = \frac{813}{27} = 30.11 \Rightarrow \text{Mean} = 30.11$$

Median = Since n = 27 (Odd)

$$\text{Median Position} = (27 + 1) / 2 = \frac{28}{2} = 14^{\text{th}} \text{ value}$$

b) Mode.

Frequency of repeated values

$$\begin{array}{lll} \rightarrow 16 \rightarrow 2 & 22 \rightarrow 2 & 33 \rightarrow 2 \\ 20 \rightarrow 2 & 25 \rightarrow 4 & 35 \rightarrow 4 \end{array}$$

The highest frequency is 4.

Mode = 25 and 35

5) Given Data (Sorted)

801:- 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215

Total records = 12

Number of Bins = 3

a) Equal-frequency Partitioning

$$\frac{12}{3} = 4 \text{ values}$$

<u>Bin</u>	<u>Values</u>
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Bin 1	5, 10, 11, 13
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Bin 2	15, 35, 50, 55
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Bin 3	72, 92, 204, 215
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b) Equal width Partitioning

Min = 5 Max = 215

$$\begin{aligned} \text{Range} &= \text{Max} - \text{Min} &= \frac{210}{3} \\ &= 215 - 5 &= 70 \\ &= \underline{210} \end{aligned}$$

→ Bin 1: 5-74 → Bin 2: 75-144 → Bin 3: 145-215

c) Clustering Partitioning.

Group similar values together.

<u>cluster</u>	<u>values</u>
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cluster 1	5, 10, 11, 13, 15
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cluster 2	35, 50, 55, 72, 92
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cluster 3	204, 215.
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e) Find Quartile Third Quartile.

sol: Lower half:

18, 19, 16, 16, 19, 20, 21, 22, 22, 25, 25, 25

Middle value = 20

Upper half:

30, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Middle value = 35

- 0 -

2
sol:

Given Data:-

18, 22, 25, 42, 28, 43, 33, 35, 36, 56, 28

Number of Observations (n) = 10

$$\text{Mean} = \frac{18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 36 + 56}{10}$$

$$= \frac{330}{10} = 33, \quad \text{Mean} = 33$$

Mean Deviation (MAD):

$$\text{MAD} = \frac{\sum |x - \bar{x}|}{n} = \frac{88}{10} = 8.8$$

Mean = 33

Mean Absolute Deviation (MAD) = 8.8

3
sol: Given Data:-

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 28, 30

33, 35, 35, 35, 36, 40, 45, 46, 52, 70

step 1: Bin depth = 3

